

Practically Implementable Minimal Universal Gate Sets for Multi-Qudit Systems with Cryptographic Validation

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Abstract. The rapid growth of quantum technologies highlights the need for scalable models of computation that go beyond qubits and exploit the richer structure of Qudits. This paper introduces a novel and efficient approach for defining universal gate sets specifically designed for higher-dimensional Qudit systems ($N \geq 2$), addressing the limitations of traditional qubit-based approaches. We present a systematic methodology for constructing fundamental Qudit gates through the inherent structure of Qudit operators, providing a robust theoretical foundation for universal quantum computation with Qudits. Our rigorously proven universal and minimal gate set enables more efficient quantum circuit design. We demonstrate the construction of multidimensional extensions of controlled operations from these fundamental elements. To facilitate practical application, we provide a Python-based algorithm for decomposing arbitrary multi-Qudit operations, accompanied by detailed time- and space-complexity analyses. The framework is further validated through end-to-end implementations of Grover’s algorithm and QKD, comparing traditional gate constructions with circuits entirely synthesized from the proposed universal gate set.¹ These validations demonstrate not only the functional equivalence of the decomposed circuits, but also their direct relevance to the advancement of cryptographic protocols, paving the way for more efficient and secure Qudit-based quantum cryptography.

Keywords: Multi-fold Qudit, Minimal Universal Quantum gates, Reck’s Decomposition, Quantum Cryptology, Grover’s algorithm, QKD, Python-Cirq.

1 Introduction

Quantum computing has so far been dominated by qubits, the familiar two-level systems that act as basic units of quantum information. Multi-state quantum systems (Qudits) offer inherent advantages over traditional qubits. These higher-dimensional units can carry more information per particle and open the door to faster computations, more compact circuits, and new directions for quantum communication and security. For quantum cryptography, this shift is especially important. Protocols such as quantum key distribution (QKD), quantum secret sharing, and secure multiparty computation could

¹ Code Repository : Minimal-Universal-Multi-Qudit-Gate-Sets

all benefit from the richer structure of Qudits. At the same time, they also raise new challenges, since building and controlling the right quantum gates becomes more complex as the system dimension grows. Qudits allow quantum circuits to be built more efficiently, making cryptographic algorithms faster, more compact, and potentially more secure. While universal gate sets for qubits are well established, extending this idea to Qudits opens up exciting opportunities for advanced quantum cryptography. In this work, we present a systematic and practical way to define fundamental Qudit gates. Using an algebraic framework, we prove that our proposed set of gates is both universal and minimal. We also validate the framework by showing that it correctly recovers the standard qubit gates as a special case. This foundation makes it possible to design more powerful and efficient quantum protocols, with clear benefits for quantum key distribution, secure multiparty computation, Grover’s search, and quantum secret sharing. Table 1 summarizes the notations used throughout the paper.

Table 1: Overview of the Notations

Notation	Description
N	Number of states or “fold” of a single Qudit system (dimension of one Qudit)
n	Number of Qudits in the system (total system dimension is N^n)
$ y\rangle$	General Qudit state vector
$ i\rangle$	Basis state vector for a Qudit
$\mathcal{H}_N$	N -dimensional complex Hilbert space
$U(N)$	Group of $N \times N$ unitary matrices
$SU(N)$	Group of $N \times N$ special unitary matrices (determinant 1)
I_N	Identity matrix of order N
E_{ij}	Matrix with 1 in the (i, j) position and zeros elsewhere
H	Hadamard gate
T	T-gate (a single-qubit phase gate with $\pi/4$ rotation)
S	S-gate (a single-qubit phase gate with $\pi/2$ rotation)
$CNOT$	Controlled-NOT gate (fundamental 2-qubit gate)
Φ	General phase shift operator (diagonal unitary matrix)
$\text{Phase}(j, \phi_j)$	Specific form of a phase gate
PHASE	Collection of general phase gates $\{\text{Phase}(j, \phi_j)\}$
PHASE_1	Subset of PHASE, specific phase gate $\{\text{Phase}(1, \theta)\}$
$T_{i,j}$	Positional subspace embedding operator for a 2×2 matrix
$T_{i,j}(U)$	Instance of $T_{i,j}$ embedding a 2×2 matrix U
$T_{elements}$	The set of $T_{i,j}(U)$ matrices
$R_{i,j}$	Rotational matrices in Reck’s decomposition
$\mathcal{S}$	The proposed optimal universal gate set ($\text{PHASE}_1 \cup T_{elements}$)
$u(m)$	Lie algebra of the unitary group $U(m)$
$su(m)$	Lie algebra of the special unitary group $SU(m)$
$\text{span}(iI_m)$	The span of imaginary identity matrices

1.1 Motivation

The pursuit of universal quantum computation, historically focused on qubit-based systems, has yielded significant insights into the structure and capabilities of quantum algorithms. However, the potential of multi-dimensional Qudit systems, with their enhanced information storage and processing capacity, remains largely untapped, especially within the cryptographic landscape. Our primary motivation stems from the recognized advantages of Qudits in various computational and cryptographic contexts. For instance, in quantum key distribution (QKD) protocols, Qudits can be realistically generated and offer the advantage of storing more information per quantum bit [24]. This capability translates to potentially higher secret key generation rates and improved resilience against eavesdropping attacks compared to traditional qubit-based QKD protocols. Furthermore, in the realm of cryptanalysis, multi-valued Grover’s search algorithms [23], which can be realized with Qudits, offer a demonstrable time advantage over their qubit-based counterparts. This opens exciting possibilities for accelerating certain cryptanalytic tasks, thereby driving the need for more sophisticated quantum-resistant cryptographic solutions. By efficiently implementing Qudit-based quantum circuits, universal Qudit gates optimize quantum-cryptographic protocols and bolster their security, while simultaneously augmenting the effectiveness of quantum attacks on classical cryptosystems through Qudits’ enhanced capabilities.

1.2 Prior-art Search

While significant strides have been made in establishing universal gate sets for qubits, such as the $\{H, T, S, CNOT\}$ set [1], the extension of these principles to higher dimensional Qudit systems has been a more complex undertaking. Early contributions by Adriano Barenco et al. [2] laid the groundwork for understanding universal quantum gate sets, exploring their construction and utility. Subsequent works, such as those by Deutsch et al. [4, 5] and A. Yu. Kitaev [6], deepened the theoretical understanding of universality for multi-qubit systems. More recent studies, including those by Laurin E. Fischer et al. [7] and Boykin et al. [9], have extended this concept to fault-tolerant systems and higher-dimensional Qudits, highlighting their potential to optimize quantum circuits and enhance computational efficiency. In parallel, efforts have been made to explore alternative perspectives on universality. For instance, works by Cafaro et al. [10] and Timothy F. Havel [11] have examined quantum gate sets through the lens of geometric algebra, while Vlasov [13] applied Clifford algebras to construct universal gates for n -qubit systems. In 2002, Shi’s work [14] demonstrated the near-universality of the Toffoli and $CNOT$ gates, further emphasizing the role of minimal gate additions in achieving universality. The study of universality of single Qudit gates [19] and semi-universality of certain subset of Qudit gates [20, 21, 22] work as a stepping stone for the study of universality of Qudit gates. Building on this extensive body of work, and being motivated by the study of universal qutrit gates by Carlos Efraim Quintero Narvaez [15], unitary matrix decomposition by Chi-Kwong Li et al. [16], and qubit gates decomposition by Juha J. Vartiainen et al. [17], our work introduces an algebraic framework for proving universality that focuses on identifying the finite, elementary operations that

are physically realizable, leading to a systematic decomposition and practical implementation of arbitrary Qudit operations. Our approach is inspired by the experimental realization of arbitrary unitary operators demonstrated by Michael Reck et al. [18], which provides a bridge between theoretical constructs and practical implementations.

From a cryptographic perspective, Qudits used in quantum key-distribution protocols [24, 39, 40] can be realistically generated and have the advantage of storing more information than qubits, and offer higher key rates over qubit based QKD. This capability translates to reduced generation time and faster computation. In view of quantum attacks on classical cryptosystems, multi-valued Grover search [23] provides a time advantage over Grover’s algorithm implemented on qubits. Notably, Qudits offer stronger security guarantee in quantum secure multiparty computation [25, 26]. Furthermore, Qudits enable increased information content per Qudit in quantum secret sharing [27, 28], demonstrating their broad applicability and potential to advance quantum technologies. Recent progress in quantum machine learning highlights the potential of Qudits over qubits, demonstrating their capacity for more intricate feature mappings [30], enhanced model expressiveness, and improved efficiency [31], ultimately enabling more robust and powerful algorithms. Prior analytical approaches have often focused on specific dimensionalities or lacked a systematic, constructive methodology for identifying minimal universal gate sets. Furthermore, the practical implementation and algorithmic decomposition of such Qudit gates have presented distinct challenges. This research addresses these gaps by providing a unified and theoretically robust framework for Qudit gate set universality.

1.3 Contribution

Unlike earlier qubit-centric studies, our formulation directly links mathematical universality to experimentally achievable operations and explicitly quantifies decomposition complexity, bridging theoretical completeness with cryptographic implementability. The key contributions are enlisted below.

- **Proof of Minimality of Universal Gate Set:** We establish a rigorous theoretical framework for universal gate sets in higher-dimensional Qudit systems (Section 3.1), introducing a robust universal set $\mathcal{S} = PHASE_1 \cup T_{elements}$. Our key contribution is the identification and explicit proof of a minimal set of indispensable gates, comprising elementary phase shifts and essential $SU(2)$ subspace embeddings. This set is proven to approximate any unitary operation on an N -dimensional Qudit system with arbitrary precision, ensuring universality. The novelty of this work lies in its Lie-algebraic proof of the minimality of the gate set proven in Theorem 3. This mathematical rigor is complemented by a detailed complexity analysis of the algorithm used to construct any arbitrary Qudit gate from this universal set.
- **Practical Validation and Applications:** The quantitative functional equivalence across arbitrary Qudit dimensions through fidelity metrics has been discussed in Section 3.4, and Section 3.5 establishes superior decomposition efficiency over a widely used alternative universal gate sets [16]. To demonstrate applicability, we

implemented the decomposition end-to-end and validated it on cryptographically relevant examples: multi-ary Grover search and a QKD workflow. For each protocol we constructed two parallel circuits: (i) the conventional circuit built from high-level Qudit gates, and (ii) a circuit in which every high-level gate is replaced by its decomposition into $T_{elements}$ and $PHASE_1$ elements following the algorithm in Section 3.2. The software implementation uses Python/Cirq primitives, so each decomposed element is directly realizable on platforms that expose only low-level rotations and phase shifts. Simulation results show functional equivalence (unstructured search amplitude amplification and matching QKD key bits) while revealing the expected trade-offs: decomposed circuits incur higher gate counts and depth but gain hardware-agnostic portability, explicit resource accounting, and finer-grained error budgeting. Quantitative validation, comparison plots and implementation notes are provided in the new Section 3.6. Collectively, these contributions deliver a minimal, implementable universal basis for Qudits together with the algorithms and empirical evidence necessary to adopt that basis in cryptographic protocols and comparative cryptanalysis.

1.4 Cryptographic Impact of the Minimal Qudit Gate Set

The identification of a minimal universal gate set for Qudits has direct implications for the design and efficiency of quantum cryptographic protocols. Qudit-based cryptography leverages higher-dimensional Hilbert spaces to enhance both information capacity and robustness against noise, but a long-standing obstacle has been the practical implementability of arbitrary Qudit unitaries within constrained hardware models. Our result addresses this bottleneck by providing a minimal, yet universal, set of gates that can be synthesized with provable efficiency.

- **Quantum Key Distribution Protocols:** Multi-level QKD schemes (e.g., using d -ary alphabets) require arbitrary basis changes and state encoding. The proposed gate set enables such transformations with reduced circuit depth, thereby lowering error rates and improving achievable key rates.
- **Quantum Cryptanalysis:** Qudit-enhanced Grover search has been shown to offer faster runtimes compared to qubit-based implementations. By providing an explicit constructive method to decompose arbitrary Qudit unitaries, our gate set strengthens the practicality of these algorithms, thereby accelerating quantum attacks on classical cryptosystems. This dual-use perspective is crucial: while our gate set empowers defensive primitives such as QKD, it also highlights the urgency of developing post-quantum cryptography resilient against Qudit-based adversaries.
- **Quantum Secret Sharing:** Secret sharing protocols based on d -ary linear codes rely on efficient encoding of quantum states into multi-Qudit subspaces. Recent constructions using symmetric entanglements and novel partitioning techniques [41, 42] demonstrate that such encoding can be efficiently realized. Our work implies that these transformations can be compiled into circuits with polynomial overhead in the Qudit dimension, making high-arity secret sharing more practical.

- **Secure Multiparty Computation:** Qudit-based MPC protocols benefit from the ability to implement controlled rotations and higher-arity entangling operations. By proving minimality, we ensure that such gates can be realized from the smallest possible universal basis, reducing both implementation complexity and fault-tolerance overhead.
- **Algorithmic Advantages:** Beyond direct protocol design, the constructive universality result facilitates the integration of higher-arity search and error-resilient primitives in cryptographic applications, where efficiency and scalability are crucial.

The proposed gate set not only advances the theoretical understanding of universality in Qudit systems but also establishes a practically implementable foundation for next-generation quantum cryptography. Its efficiency and minimality strengthen the applicability of Qudit-based designs in QKD, secret sharing, and multiparty computation, while clarifying their implications for quantum cryptanalysis.

2 Definition and Prerequisites

To facilitate understanding of the proposed methods, this section provides the essential definitions and mathematical prerequisites relevant to the study. These concepts serve as the basis for the development and analysis presented in the following sections.

2.1 Subspace Embedding

A subspace embedding is a technique in linear algebra that maps a lower-dimensional subspace, spanned by a set of vectors or matrices, into a higher-dimensional space. This mapping preserves the essential structure and properties of the original subspace while satisfying specific conditions such as distance preservation or bounded distortion. Consider $T_{i_1, i_2, i_3}(U)$ that embeds a 3×3 matrix U into a 6×6 identity matrix:

$$U = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \implies T_{0,1,2}(U) = \begin{pmatrix} a_{11} & a_{12} & a_{13} & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

2.2 Representation of Qudit

In quantum mechanics, qubits are two-dimensional quantum systems represented by the basis states $\{|0\rangle, |1\rangle\}$. While qubits form the foundation of many quantum computing architectures, the concept can be generalized to higher-dimensional systems known as Qudits [34]. Mathematically, a Qudit is a unit vector in an N -dimensional complex Hilbert space, $\mathcal{H}_N$. This space is spanned by an orthonormal basis of N states, labeled as $|0\rangle, |1\rangle, |2\rangle, \dots, |N-1\rangle$. A general Qudit state $|\psi\rangle$, is a linear combination of these

basis states $|\psi\rangle = \sum_{i=0}^{N-1} c_i |i\rangle$, where $c_i \in \mathbb{C}$ are complex coefficients satisfying the normalization condition $\sum_{i=0}^{N-1} |c_i|^2 = 1$. In its simplest representation, an N -dimensional Qudit can be visualized as a column vector $|\psi\rangle = (c_0, c_1, \dots, c_{N-1})^T$, where each c_i corresponds to the amplitude of the i -th basis state, and the superscription T represents the transpose of a vector. Using the orthonormal basis vectors $|i\rangle$, this can be expressed as $|\psi\rangle = c_0 (1, 0, \dots, 0)^T + c_1 (0, 1, \dots, 0)^T + \dots + c_{N-1} (0, 0, \dots, 1)^T$. Here, each basis state $|i\rangle$ is represented as a column vector with 1 in the i -th position and 0 elsewhere: $|i\rangle = (0, 0, \dots, 0, 1, 0, \dots, 0)^T$. This method emphasizes that Qudits provide a versatile way to represent quantum systems, offering more possibilities than qubits. Thus, the set $\{|0\rangle, |1\rangle, |2\rangle, \dots, |N-1\rangle\}$ serves as the orthonormal basis for all N -dimensional Qudits.

2.3 Representation of Qudit Gate

Qudits, generalizations of qubits representing quantum states in a N -dimensional Hilbert space, are manipulated using *Qudit gates*, which are unitary operators acting on these N -dimensional spaces [29, 32, 37]. A single-Qudit gate is represented by an $N \times N$ unitary matrix, $U \in U(N)$, where $UU^\dagger = I_{N \times N}$ ($I_{N \times N}$ being the $N \times N$ identity matrix). For example, a single-Qudit gate acting on a three-level qutrit ($N = 3$) might be represented by the following unitary matrix:

$$U = \begin{pmatrix} e^{i\pi/4} & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ 0 & -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

This unitary matrix ensures $UU^\dagger = I_{3 \times 3}$, preserving the probabilistic interpretation of quantum mechanics. Multi-Qudit gates extend this concept to systems involving multiple Qudits, where a gate acting on an N^n -dimensional system is represented by a $N^n \times N^n$ unitary matrix. For example, a two-qutrit gate ($N = 3$, $n = 2$) is described by a 9×9 matrix.

Subset of $SU(N^n)$ Consider a 2×2 matrix $U = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} \in SU(2)$. We can embed this matrix into a $N^n \times N^n$ matrix $T_{i,j}$ in the positions i and j as

$$T_{i,j}(U) = \begin{pmatrix} 1 & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_{11} & \dots & u_{12} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_{21} & \dots & u_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 0 & \dots & 1 \end{pmatrix}$$

We define $T_{elements} = \{T_{i,j}(U) | 0 \leq i < j < N^n, U \in SU(2)\} \subset SU(N^n)$ as the set of $T_{i,j}$.

PHASE Qudit Gates Diagonal unitary matrices, where all diagonal elements have a modulus of 1, can be expressed as:

$$\Phi = \begin{pmatrix} e^{i\phi_0} & 0 & \dots & 0 \\ 0 & e^{i\phi_1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{i\phi_{N^n-1}} \end{pmatrix}.$$

These matrices can be systematically constructed using a specific form:

$$Phase(j, \phi_j) = \begin{pmatrix} 1 & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & e^{i\phi_j} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & 1 \end{pmatrix},$$

where j varies from 0 to $N^n - 1$, and ϕ_j varies over $[-\pi, \pi)$. For Qudits of N -fold, n -dimensional, the matrix $Phase(j, \phi_j)$ will have order $N^n \times N^n$. The collection of these matrices is defined as: $PHASE = \{Phase(j, \phi_j) | j \in \{0, 1, \dots, N^n - 1\}, \phi_i \in [-\pi, \pi)\}$. This set is capable of generating any phase-shift operator [35] for an n -dimensional N -fold Qudit gate. We now define a special subset $PHASE_1 \subset PHASE$ as follows.

$$PHASE_1 = \left\{ \begin{pmatrix} e^{i\theta} & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \text{ where } \theta \in [-\pi, \pi) \right\} = \{Phase(1, \theta) | \text{where } \phi_i \in [-\pi, \pi)\}$$

2.4 Decomposition of Unitary Matrices with Reck's Approach

Michael Reck et al. [18] demonstrated that any $N \times N$ unitary matrix A can be reduced to a diagonal matrix D with elements of modulus 1 by sequentially applying a series of $T_{i,j}$ transformations. These transformations are derived from $N \times N$ identity matrices, where the elements at positions $(i, i), (i, j), (j, i), (j, j)$ are replaced by the corresponding elements of a 2×2 unitary matrix $U_{i,j} \in U(2)$. Conceptually, the $T_{i,j}$ matrices are embeddings of $U_{i,j}$ into an N -dimensional subspace. For simplicity, we will denote $T_{i,j}(U_{i,j}) = T_{i,j}$. The resulting diagonal matrix D , being unitary and having diagonal elements of modulus 1, must take the form:

$$D = \begin{pmatrix} e^{i\theta_1} & 0 & \dots & 0 \\ 0 & e^{i\theta_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{i\theta_N} \end{pmatrix}_{N \times N},$$

where e^{ik} represents a phase factor. Its inverse, also a diagonal unitary matrix, is given by:

$$D^{-1} = \begin{pmatrix} e^{-i\theta_1} & 0 & \dots & 0 \\ 0 & e^{-i\theta_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{-i\theta_N} \end{pmatrix}_{N \times N} = \begin{pmatrix} e^{i(-\theta_1)} & 0 & \dots & 0 \\ 0 & e^{i(-\theta_2)} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{i(-\theta_N)} \end{pmatrix}_{N \times N}.$$

The process of sequentially applying $T_{i,j}$ transformations can be described mathematically as follows: $A \cdot T_{N,N-1} \cdot T_{N,N-2} \cdots T_{N,1} \cdot T_{N-1,N-2} \cdots T_{2,1} = D \implies A = D \cdot T_{2,1}^{-1} \cdot T_{3,1}^{-1} \cdot T_{3,2}^{-1} \cdots T_{N,N-1}^{-1}$. Rewriting in terms of $D^{-1} = E$, where E is a diagonal unitary matrix: $A = (T_{N,N-1} \cdot T_{N,N-2} \cdots T_{2,1} \cdot E)^{-1}$. Since E is diagonal with all elements of modulus 1, the inverse of A is also unitary. Now, consider an arbitrary unitary matrix M . Its inverse $M^{-1} = A$, can be decomposed using the above result: $M^{-1} = A = (T_{N,N-1} \cdot T_{N,N-2} \cdots T_{2,1} \cdot E)^{-1}$. Therefore, the unitary matrix M can be expressed as: $M = T_{N,N-1} \cdot T_{N,N-2} \cdots T_{2,1} \cdot E$. In general, any $N \times N$ unitary matrix M can be decomposed as: $\prod_{i,j=0,j < i}^{N-1} R_{i,j} \cdot \Phi$, where $R_{i,j} = T_{i,j}(U_{i,j})$ for some $U_{i,j} \in SU(2)$ and Φ is an $N \times N$ phase shift matrix. We can state the following theorem that describes the aforementioned decomposition.

Theorem 1. (Reck's Decomposition[18]) Any unitary matrix of order N can be decomposed into two-dimensional subspace rotations and a phase shift Φ as follows $M = \prod_{i,j=0,j < i}^{N-1} R_{i,j} \cdot \Phi$, where $R_{i,j} = T_{i,j}(U_{i,j})$ for some $U_{i,j} \in SU(2)$.

3 Proposed Optimal Universal Gate Set

In this section, a novel methodology for achieving universality with sets of N -fold n -dimensional qubit gates is presented. Unlike prior quantum gate synthesis methods, this framework directly supports N -valued qubit gates, ensuring universal computation with optimality of resources. The core of the protocol involves multiplication of Qudit gates, which enables constructing arbitrary unitary operations from a finite proposed gate set.

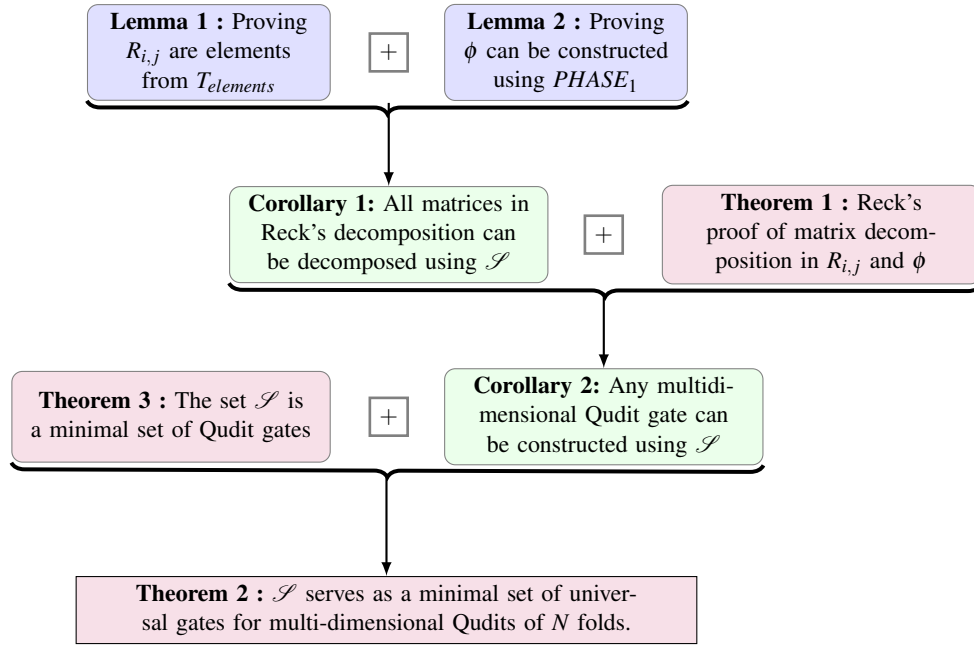
3.1 Universality for Qudit Gates

We define $\mathcal{S} = PHASE_1 \cup T_{elements}$ and present the following Theorem 2, which we will prove in the subsequent section.

Theorem 2. The set $\mathcal{S} = PHASE_1 \cup T_{elements}$ serves as a minimal set of universal gates for multi-dimensional Qudits of N folds.

To prove the Theorem 2, we will employ Theorem 1, along with Lemma 1, Lemma 2, Corollary 2 and Theorem 3. The flow Diagram 1 depicts the Flow of the proof.

Lemma 1. Each matrix $R_{i,j}$ in the Reck's decomposition (Theorem 1) is an element of the set $\mathcal{T}_{elements}$.



Flow Diagram 1: Process flow of the proof of Theorem 2

Proof. In Reck's decomposition, we express M as $M = \prod_{i,j=0, j < i}^{N-1} R_{i,j} \Phi$, where $R_{i,j} = T_{i,j}(U_{i,j})$ for some $U_{i,j} \in SU(2)$.

From definition of the set $T_{elements} = \{T_{i,j}(U) | 0 \leq i < j < N^n, U \in SU(2)\}$, we see that each of the $T_{i,j}$ matrices belong to $T_{elements}$ itself. Also, from Reck's decomposition, $R_{i,j} = T_{i,j}(U)$ where $U \in SU(2)$. Thus, each of the $R_{i,j}$ can be constructed with $T_{elements}$.

Lemma 2. Any phase shift matrix Φ in the Reck's decomposition (Theorem 1) can be constructed from elements of $PHASE_1 = \{Phase(1, \phi) | \text{where } \phi_i \in [-\pi, \pi]\}$ and $T_{elements}$.

Proof. Let us consider the phase shift matrix Φ of size $N^n \times N^n$ defined as:

$$\Phi = \begin{pmatrix} e^{i\phi_0} & 0 & 0 & \dots & 0 \\ 0 & e^{i\phi_1} & 0 & \dots & 0 \\ 0 & 0 & e^{i\phi_2} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & e^{i\phi_{N^n-1}} \end{pmatrix}.$$

This matrix Φ is a diagonal matrix, where the diagonal entries are complex exponential, $e^{i\phi_0}, e^{i\phi_1}, e^{i\phi_2}, \dots, e^{i\phi_{N^n-1}}$ representing phase shifts. We want to show that Φ can be

represented as a product of elements from $PHASE = \{Phase(i, \phi_i) \mid i \in \{0, 1, \dots, N^n - 1\}, \text{ where } \phi_i \in [-\pi, \pi)\}$. Observe that the phase matrix Φ can be expressed as of the form:

$$\Phi = \begin{pmatrix} e^{i\phi_0} & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & e^{i\phi_1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \dots \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{i\phi_{N^n-1}} \end{pmatrix}$$

$$\implies \Phi = Phase(0, \phi_0)Phase(1, \phi_1) \dots Phase(N^n - 1, \phi_{N^n-1}).$$

So, each of these matrices belongs to the set $PHASE = \{Phase(i, \phi_i) \mid i \in \{0, 2, \dots, N^n - 1\}, \phi_i \in [-\pi, \pi)\}$. Now, a matrix $Phase(j, \theta_j)$ from $PHASE$ can be decomposed using element from $PHASE_1$ and $T_{elements}$ as follows.

$$\begin{pmatrix} 1 & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & e^{i\phi_j} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & 1 \end{pmatrix} = \begin{pmatrix} e^{i(-\phi_j)} & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & e^{i\phi_j} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & 1 \end{pmatrix} \times \begin{pmatrix} e^{i\phi_j} & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

Here the first matrix in the product is a member of $T_{elements}$, and the second one is from $PHASE_1 \subset PHASE$, defined above. Thus any matrix in the decomposition $\Phi = Phase(0, \phi_0)Phase(1, \phi_1) \dots Phase(N^n - 1, \phi_{N^n-1})$ can be constructed with elements from $PHASE_1$ and $T_{elements}$, ultimately leading to any phase shift operator matrix Φ can be constructed using $PHASE_1$ and $T_{elements}$ sets only. This completes the proof.

Corollary 1. *All the matrices in Reck's decomposition (Theorem 1) can be decomposed using matrices of the set $\mathcal{S}$.*

Proof. The two types of matrices in the expansion of Reck's decomposition are $R_{i,j}$ and Φ . From Lemma 1, $R_{i,j}$ can be constructed using $T_{elements} \subset \mathcal{S}$. Also, from Lemma 2, ϕ can be constructed using $PHASE_1 \subset \mathcal{S}$. Thus, any matrix in expansion of Reck's decomposition, can be decomposed using $\mathcal{S}$.

Corollary 2. *The set $\mathcal{S} = PHASE_1 \cup T_{elements}$ forms a universal gate set for multidimensional Qudit gates of N -fold dimension.*

Proof. Any unitary matrix can be decomposed according to Reck's decomposition, with the $R_{i,j}$ and Φ matrices only (Theorem 1). Since any multidimensional Qudit gate of N -fold dimension is an N^n -order unitary matrix, it follows that such gates can be decomposed as a product of elements from $\mathcal{S} = PHASE_1 \cup T_{elements}$. This demonstrates that the set $\mathcal{S}$ is universal for constructing all multidimensional Qudit gates of N -fold dimension.

Theorem 3. *The set $\mathcal{S} = PHASE_1 \cup T_{elements}$ forms a minimal set of universal gates for multidimensional Qudit gates of N -fold and n -dimension.*

Proof. Let $\mathcal{S} = PHASE_1 \cup T_{elements}$ be a gate set intended for the generation of the unitary group $U(N^n)$, for positive integers N and n . We aim to formally demonstrate that both $PHASE_1$ and $T_{elements}$ are indispensable components of $\mathcal{S}$ for achieving universality in $U(N^n)$. For this, we rely on the distinct properties of the Lie algebras [43] generated by each subset. The Lie group generated by a set of gates G is denoted by $Lie(G)$, and its associated Lie algebra by $LieAlg(G)$.

The Lie algebra of $U(N^n)$, denoted $u(N^n)$, consists of all $N^n \times N^n$ anti-Hermitian matrices. It admits a direct sum decomposition $u(N^n) = su(N^n) \oplus \text{span}(iI_{N^n})$ where $su(N^n)$ is the Lie algebra of the special unitary group $SU(N^n)$ (comprising trace-zero anti-Hermitian matrices), and $\text{span}(iI_{N^n})$ represents the one-dimensional subalgebra associated with global phase changes in $U(N^n)$. Our argument demonstrates that $T_{elements}$ is essential for spanning $su(N^n)$ and $PHASE_1$ for spanning $\text{span}(iI_{N^n})$.

1. Necessity of $T_{elements}$ for Generating $su(N^n)$ (Mixing Operations)

- Definition and Lie Algebra of $PHASE_1$: The set $PHASE_1$ is defined as composed of individual gates of the form $\text{Phase}(1, \theta) = \text{diag}(e^{i\theta}, 1, \dots, 1) \in U(M)$ for $\theta \in [-\pi, \pi]$. The infinitesimal generator associated with $\text{Phase}(1, \theta)$ is obtained by taking the derivative with respect to θ at $\theta = 0$:

$$L(PHASE_1) = \frac{d}{d\theta} \text{diag}(e^{i\theta}, 1, \dots, 1) \Big|_{(\theta=0)} = i \cdot \text{diag}(1, 0, \dots, 0) = iE_{11}$$

where E_{11} is the matrix unit with a 1 in the (1,1) position and zeros elsewhere. The Lie algebra $LieAlg(PHASE_1)$ generated by $PHASE_1$ is therefore $\text{span}(iE_{11})$.

- Properties of $LieAlg(PHASE_1)$: Any element in $LieAlg(PHASE_1)$ is a diagonal anti-Hermitian matrix. Furthermore, the set of all diagonal $N^n \times N^n$ anti-Hermitian matrices forms a Lie subalgebra of $u(N^n)$. Any sum or Lie bracket (commutator) of diagonal matrices yields a diagonal matrix. Consequently, the Lie algebra $LieAlg(PHASE_1)$ contains only diagonal matrices.
- Inability of $PHASE_1$ to Generate $su(N^n)$'s Non-Diagonal Components: The Lie algebra $su(N^n)$ for $N \geq 2$, $n \geq 1$ contains numerous non-diagonal anti-Hermitian matrices. For instance, in $su(2)$, the generators $i\sigma_x = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ and $i\sigma_y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ are non-diagonal. Since $LieAlg(PHASE_1)$ consists exclusively of diagonal matrices, it cannot generate these non-diagonal elements of $su(N^n)$. Therefore, the Lie group $Lie(PHASE_1)$ cannot generate arbitrary 'mixing' operations, which are represented by non-diagonal elements in $SU(N^n)$.
- Role of $T_{elements}$ in Generating $su(N^n)$: The set $T_{elements}$ consists of $U(N^n)$ gates that are embeddings of $SU(2)$ matrices. The Lie algebra $su(2)$ contains non-diagonal trace-zero anti-Hermitian matrices ($i\sigma_x, i\sigma_y, i\sigma_z$). A fundamental result in quantum control and universal gate sets (e.g., related to Reck's

decomposition and the structure of $SU(N^n)$ establishes that $SU(N^n)$ can be generated by such embedded $SU(2)$ operations. This implies that the Lie algebra $\text{LieAlg}(T_{\text{elements}})$ can generate the necessary non-diagonal elements to span $su(N^n)$.

- Conclusion on T_{elements} 's Necessity: Without T_{elements} , the gate set $\mathcal{S}$ would effectively be reduced to PHASE_1 . As shown, $\text{LieAlg}(\text{PHASE}_1)$ is a proper sub-algebra of $u(N^n)$, specifically lacking the ability to generate non-diagonal transformations in $su(N^n)$. Such transformations are critical for state mixing, entanglement, and spanning $SU(N^n)$. Thus, T_{elements} is indispensable for $\mathcal{S}$ to achieve universality for $U(N^n)$.

2. Necessity of PHASE_1 for Generating $\text{span}(iI_{N^n})$ (Global Phase Operations)

- Properties and Lie Algebra of T_{elements} : Each gate $T \in T_{\text{elements}}$ is an embedding of an $SU(2)$ matrix into $U(N^n)$. By definition, all $SU(2)$ matrices have a determinant of 1. Since the determinant is a multiplicative homomorphism from a Lie group to $\mathbb{C}^*$, any product of gates from T_{elements} (and thus any element in $\text{Lie}(T_{\text{elements}})$) must also have a determinant of 1. This implies that $\text{Lie}(T_{\text{elements}}) \subseteq SU(N^n)$. The Lie algebra $\text{LieAlg}(T_{\text{elements}})$ is a subalgebra of $su(N^n)$. By definition, all elements of $su(N^n)$ are trace-zero anti-Hermitian matrices. Therefore, any element $L \in \text{LieAlg}(T_{\text{elements}})$ must satisfy $\text{Tr}(L) = 0$.
- Inability of T_{elements} to Generate $\text{span}(iI_{N^n})$: The $u(N^n)$ Lie algebra includes elements of the form ixI_{N^n} for $x \in [-\pi, \pi]$. The trace of such a matrix is $\text{Tr}(ixI_{N^n}) = iN^n x$. For $N, n > 0$ and $x \neq 0$, this trace is non-zero. Since all elements in $\text{LieAlg}(T_{\text{elements}})$ are trace-zero, $\text{LieAlg}(T_{\text{elements}})$ cannot generate any element of the form ixI_{N^n} where $x \neq 0$. Consequently, the Lie group $\text{Lie}(T_{\text{elements}})$ cannot generate operations that effect a non-trivial global phase change (i.e., elements in $U(N^n) \setminus SU(N^n)$).
- Role of PHASE_1 in Generating $\text{span}(iI_{N^n})$: The gate $\text{Phase}(1, \theta)$ defined as $\text{Phase}(1, \theta) = \text{diag}(e^{i\theta}, 1, \dots, 1) \in \text{PHASE}_1$ has an infinitesimal generator iE_{11} . This generator has a non-zero trace, specifically $\text{Tr}(iE_{11}) = i$. This demonstrates that $\text{LieAlg}(\text{PHASE}_1)$ contains elements with a non-zero trace. Importantly, the parameter θ in PHASE_1 is continuous ($\theta \in [-\pi, \pi]$). This continuous degree of freedom is essential for generating a continuous range of global phase shifts, which is precisely what the $U(1)$ factor in $U(N^n)$ requires (generated by iI_{N^n}). While iE_{11} itself is not iI_{N^n} , it provides the capability to introduce non-zero trace components, which are absent in $\text{LieAlg}(T_{\text{elements}})$. When combined with the $su(N^n)$ generators from T_{elements} , the non-zero trace component provided by PHASE_1 allows for the full $u(N^n)$ Lie algebra to be spanned, including the $\text{span}(iI_{N^n})$ part.
- Conclusion on PHASE_1 's Necessity: If PHASE_1 were removed, the gate set $\mathcal{S}$ would be reduced to T_{elements} . As shown, $\text{LieAlg}(T_{\text{elements}})$ is a proper sub-algebra of $u(M)$, specifically lacking the ability to generate operations that induce a global phase change (i.e., elements in $\text{span}(iI_{N^n})$). Such global phase changes are necessary to span the entirety of $U(N^n)$. Thus, PHASE_1 is indispensable for $\mathcal{S}$ to achieve universality for $U(N^n)$.

3. Indispensability of both $T_{elements}$ and $Phase_1$ set of matrices

The Lie algebra of the target group $U(N^n)$ decomposes precisely into two orthogonal components: $u(N^n) = su(N^n) \oplus \text{span}(iI_{N^n})$. The gate set $T_{elements}$ generates a Lie algebra $\text{LieAlg}(T_{elements})$ that is a subalgebra of $su(N^n)$. Consequently, all infinitesimal transformations generated by $T_{elements}$ are trace-zero and thus cannot contribute to the $\text{span}(iI_{N^n})$ component of $u(N^n)$.

The gate set $PHASE_1$ generates a Lie algebra $\text{LieAlg}(PHASE_1)$ that is a subalgebra of the diagonal matrices in $u(N^n)$. While $\text{LieAlg}(PHASE_1)$ cannot generate non-diagonal transformations (arbitrary elements of $su(M)$), it provides generators with non-zero traces (e.g., iE_{11}) and a continuous parameter space that enables the generation of the $\text{span}(iI_{N^n})$ component, which $T_{elements}$ cannot.

Since $\text{LieAlg}(T_{elements})$ is confined to trace-zero matrices, and $\text{LieAlg}(PHASE_1)$ is confined to diagonal matrices (and cannot span all of $su(N^n)$), neither set alone can generate the entirety of $u(N^n)$. However, their union $\mathcal{S} = PHASE_1 \cup T_{elements}$ (or, more precisely, the Lie algebra $\text{LieAlg}(\mathcal{S})$ generated by $\mathcal{S}$) is capable of spanning $u(N^n)$, as implied by the combination of established theoretical results like Reck's decomposition and the properties of phase gates in quantum optics.

Because the Lie algebras generated by $PHASE_1$ and $T_{elements}$ provide distinct and non-overlapping necessary components to span $u(N^n)$, neither $PHASE_1$ nor $T_{elements}$ can be removed from $\mathcal{S}$ without losing the ability to generate the full unitary group $U(N^n)$. This rigorously establishes the indispensability of both $PHASE_1$ and $T_{elements}$ for the universality of $\mathcal{S}$.

Proof of Theorem 2

Proof. From Corollary 2, we have that $\mathcal{S}$ is a universal gate set for any multidimensional N -fold Qudit gates. And Theorem 3 proves the minimality of the set $\mathcal{S}$ in generating any multidimensional N -fold Qudit gate. Collating Corollary 2 and Theorem 3, we have that $\mathcal{S}$ is a minimal universal gate set. This proves Theorem 2.

3.2 Algorithm for Obtaining Decomposition Matrices

We have obtained the decomposition Algorithm 3.2 for Reck's decomposition of unitary matrices. Alongside, we implemented the decomposition algorithm in a programmatic way, analyzing its space and time complexity in average case. For the sake of simplicity, we have considered the matrices' order to be N_1 instead of N^n .

This algorithm expresses an $N_1 \times N_1$ unitary matrix M as a product of $L = N_1(N_1 - 1)/2$ elementary unitary matrices R_k and a final diagonal unitary matrix Φ . The decomposition takes the form $M = R_1 \otimes R_2 \otimes \dots \otimes R_L \otimes \Phi$. Each R_k matrix is an $N_1 \times N_1$ unitary matrix, differing from the identity matrix I_{N_1} only in a 2×2 principal submatrix. These R_k matrices are analogous to the T_{pq} matrices in the original Reck's paper, representing complex Givens rotations. The algorithm iteratively transforms the input matrix M into a diagonal matrix Φ by sequentially pre-multiplying it with the Hermitian conjugates of the desired R_k matrices.

Time Complexity Analysis :

1. **Input Validation Subtotal:** $O(N_1^3)$ (dominated by $N_1 \times N_1$ matrix multiplication of $M @ \dots$).
2. **Initialization Subtotal:** $O(N_1^2)$ $U_current = M.astype(complex)$ involves creating a copy of the $N_1 \times N_1$ input matrix.).
3. **Main Loops (Givens Rotations Application):** The outer loop runs $N_1 - 1$ times. The inner loop runs $N_1 - 1 - j$ times for each j . The total number of times the inner loop body executes is $O(N_1^2)$. $U_current = R_inv_matrix @ U_current$. This is a multiplication of two $N_1 \times N_1$ matrices, leading to an $O(N_1^3)$ complexity. With $O(N_1^2)$ iterations, each containing an $O(N_1^3)$ dominant matrix multiplication, the total for the main loops is $O(N_1^2) \times O(N_1^3) = O(N_1^5)$.
4. **Final Steps:** $\Phi = U_current$ is an $O(1)$ reference assignment. Obtaining Φ in form of $PHASE_1$ matrix, consists of N_1 many multiplications, making the whole calculation $O(N_1)$.
5. **Overall Time Complexity:** $O(N_1^3)$ (validation) + $O(N_1^2)$ (initialization) + $O(N_1^5)$ (main loops) + $O(N_1)$ (final steps). The highest order term is $O(N_1^5)$. Figure 1 shows python-based validation result of time complexity analysis.

Space Complexity Analysis :

1. **Input and Initial Copies:** M : and $U_current$ both occupies $O(N_1^2)$ space.
2. **Temporary Matrices within Iterative Loops:** G_block is a temporary 2×2 matrix, consuming $O(1)$ constant space. R_inv_matrix is a temporary $N_1 \times N_1$. This matrix is allocated $O(N_1^2)$ space, and this memory is effectively reused across successive loop iterations, preventing cumulative memory growth.
3. **Stored $R_matrices$ List:** The list dynamically stores a collection of matrices. The number of matrices accumulated within this list is approximately $\frac{N_1(N_1-1)}{2}$, which asymptotically translates to $O(N_1^2)$ matrices. Each individual matrix stored in this list is an $N_1 \times N_1$ matrix, thus demanding $O(N_1^2)$ memory. The total space consumed by the $R_matrices$ list is the product of the number of stored matrices and the space required per matrix: $O(N_1^2) \times O(N_1^2) = O(N_1^4)$.
4. **Final Output Φ :** Represents the final output. It is characterized as a reference to $U_current$. Additionally, the creation of $T|elements$ matrix $\Phi_balance$ requires additional independent memory allocation proportional to N_1^2 .
5. **Overall Space Complexity:** The predominant term in the algorithm's space complexity is $O(N_1^4)$, primarily due to the storage requirements of the $R_matrices$ list. Figure 1 indicates space complexity analysis.

Algorithm 1 Reck's Unitary Matrix Decomposition

Require: $M \in \mathbb{U}^{N_1 \times N_1}$: An $N_1 \times N_1$ unitary matrix.

Ensure: R_{matrices} : An ordered list $[R_1, R_2, \dots, R_L]$ of $N_1 \times N_1$ unitary matrices.

Ensure: $\Phi \in \mathbb{U}^{N_1 \times N_1}$: A diagonal $N_1 \times N_1$ unitary matrix with all diagonal elements having a modulus of 1.

```

1:  $\epsilon \leftarrow 10^{-9}$   $\triangleright$  Numerical tolerance for checking approximate zero values
2: Initialize:
3:  $R_{\text{matrices}} \leftarrow []$   $\triangleright$  An ordered list to store the  $R_k$  matrices
4:  $U_{\text{current}} \leftarrow M$   $\triangleright$  The working matrix, which will be iteratively transformed into  $\Phi$ 

5: Iterative Annihilation (Column by Column, from bottom up):
6: for  $j$  from 0 to  $N_1 - 2$  do  $\triangleright$  Column index for the pivot element  $U_{\text{current}}[j, j]$ 
7:   for  $i$  from  $N_1 - 1$  down to  $j + 1$  do  $\triangleright$  Row index for  $U_{\text{current}}[i, j]$  to be annihilated
8:      $u \leftarrow U_{\text{current}}[j, j]$   $\triangleright$  The pivot element
9:      $v \leftarrow U_{\text{current}}[i, j]$   $\triangleright$  The off-diagonal element to be zeroed
10:    if  $|v| < \epsilon$  then
11:      continue  $\triangleright$  Element is already numerically zero, skip transformation
12:
13:    Construct  $2 \times 2$  Complex Givens Rotation Block ( $G_{\text{block}}$ ):
14:     $r \leftarrow \sqrt{|u|^2 + |v|^2}$ 
15:     $c \leftarrow \bar{u}/r$ 
16:     $s \leftarrow \bar{v}/r$ 
17:     $G_{\text{block}} = \begin{pmatrix} c & s \\ -\bar{s} & \bar{c} \end{pmatrix}$   $\triangleright$  This matrix is unitary
18:
19:    Construct Full  $N_1 \times N_1$  Inverse Transformation Matrix ( $R_{\text{inv\_matrix}}$ ):
20:     $R_{\text{inv\_matrix}} \leftarrow I_{N_1}$   $\triangleright$  Initialize as the  $N_1 \times N_1$  identity matrix
21:     $R_{\text{inv\_matrix}}[j, j] \leftarrow c, R_{\text{inv\_matrix}}[j, i] \leftarrow s$ 
22:     $R_{\text{inv\_matrix}}[i, j] \leftarrow -\bar{s}, R_{\text{inv\_matrix}}[i, i] \leftarrow \bar{c}$ 
23:
24:    Apply Transformation and Store  $R_k$ :
25:     $U_{\text{current}} \leftarrow R_{\text{inv\_matrix}} \otimes U_{\text{current}}$   $\triangleright$  This operation annihilates  $U_{\text{current}}[i, j]$ 
26:     $R_k \leftarrow R_{\text{inv\_matrix}}^H$   $\triangleright R_k$  is the Hermitian conjugate of the applied transformation
27:    Append  $R_k$  to  $R_{\text{matrices}}$ 
28:
29: Final Diagonal Matrix:
30:  $\Phi \leftarrow U_{\text{current}}$   $\triangleright U_{\text{current}}$  is now an upper triangular unitary matrix, implying it is diagonal
31: if  $|\Phi| \neq 1$  then
32:    $\Phi_{\text{balance}} \leftarrow I_{N_1}$   $\triangleright T_{\text{elements}}$  matrix to prepare PHASE1 matrix from  $\Phi$ 
33:    $v = 1$ 
34:   for  $i$  from 0 to  $N_1 - 1$  do
35:      $v = v \cdot \Phi[i, i]$ 
36:      $\Phi_{\text{balance}}[i, i] \leftarrow \Phi[i, i]$ 
37:    $\Phi_{\text{balance}}[i, i] \leftarrow \bar{v}$ 
38:   Append  $\Phi_{\text{balance}}$  to  $R_{\text{matrices}}$ 
39:    $\Phi \leftarrow I_{N_1}$ 
40:    $\Phi[0, 0] \leftarrow v$ 
41: return  $R_{\text{matrices}}, \Phi$ 

```

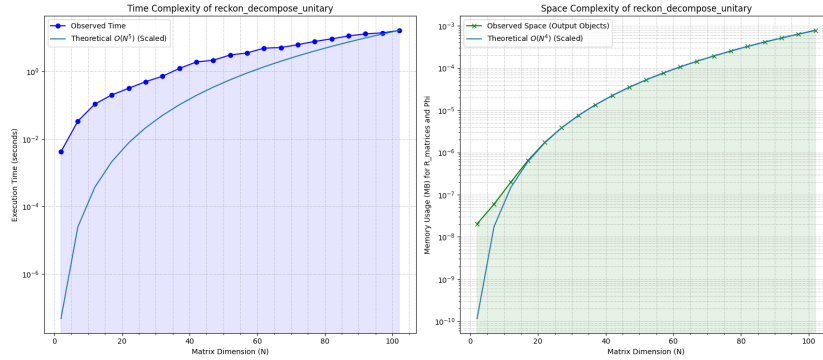



Figure 1: Average Case Time and Space Complexity Graphs for the Algorithm compared with $O(N_1^5)$ and $O(N_1^4)$ graphs respectively converted to logarithmic scale. Graphs obtained while running for various input sizes (N_1 ranging from 2 to 100+)

3.3 Construction of Existent Universal Gate Sets through Proposed Gate Sets

By Solovay-Kitaev theorem [1], it is established that for single qubit ($N = 2$), the set of gates $\mathcal{A} = \{H, T, S, CNOT\}$ constitutes a universal set of quantum gates. In our hypothesis, we propose that the set $\mathcal{S} = \{PHASE, SU(2^n)\}$ serves as a universal gate set for the case $N = 2$. To validate this claim, we demonstrate that the set $\mathcal{A}$ can be constructed entirely using gates from $\mathcal{S}$. The derivations are outlined as follows:

- For $N = 2$ and $n = 1$, the Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ can be expressed as following, where the first two matrices are from $T_{elements}$, and the last is from $PHASE_1$.

$$H = \left(\begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \times \begin{pmatrix} -1 - i6.1e-17 & 0 \\ 0 & -1 + i6.12e-17 \end{pmatrix} \right) \times \begin{pmatrix} -1 + i6.1e-17 & 0 \\ 0 & 1 \end{pmatrix}$$

- $T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$ gate can be expressed as: $T = \begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \times \begin{pmatrix} e^{i\pi/4} & 0 \\ 0 & 1 \end{pmatrix}$.
- $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$ gate can be decomposed as: $S = \begin{pmatrix} e^{-i\pi/2} & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} \times \begin{pmatrix} e^{i\pi/2} & 0 \\ 0 & 1 \end{pmatrix}$
- The Controlled-NOT ($CNOT$) gate for $N = 2$ can be decomposed into product of matrices from $T_{elements}$ and $PHASE_1$.

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \left(\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \times \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \right) \times \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

These derivations confirm that all elements of the universal gate set $\mathcal{A}$ can be expressed using the gates of $\mathcal{S}$ in the particular case of $N = 2$. Thus, for the case of $N = 2$, $\mathcal{S}$ serves as a universal gate set for single-qubit operations.

3.4 Functional Equivalence of the Proposed Universal gate sets

To quantitatively validate that the decomposed gates preserve the functional behavior of their original unitary counterparts, we conducted numerical evaluations on randomly generated Qudit unitaries. For some randomly selected dimensions between 2 and 20, ten random unitaries were generated and decomposed into the sequence of rotation matrices (R_i) and phase matrices (Φ) obtained from the proposed minimal universal set $S = \text{PHASE}_1 \cup T_{\text{elements}}$. The reconstructed matrices were then compared with the original unitaries using multiple fidelity and distance metrics widely employed in quantum verification. The following parameters were computed for every trial:

- **Operator fidelity:** $F = \frac{1}{m} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{approx}})|$, indicating global unitary overlap.
- **Trace similarity:** Normalized trace overlap between the two unitaries.
- **Frobenius error:** $\|U_{\text{target}} - U_{\text{approx}}\|_F$, quantifying element-wise deviation.
- **Spectral norm error:** $\|U_{\text{target}} - U_{\text{approx}}\|_2$, the largest singular-value deviation.
- **Eigenphase minimal arc:** Maximum phase-angle spread between eigenvalues, measured in radians.

Each metric was averaged over 10 independent trials for each Qudit dimension. All values demonstrated near-perfect agreement between the original and reconstructed matrices, with fidelity and trace-similarity consistently equal to 1.0 (up to floating-point precision), and Frobenius and spectral errors on the order of 10^{-15} . This confirms that the decomposition procedure preserves functional equivalence across dimensions. The results are summarized in Table 2.

Table 2: Functional Equivalence Validation for Random Qudit Unitaries

Qudit States	Trials Count	Operator Fidelity	Trace Similarity	Frobenius Error	Spectral Norm Error	Eigenphase Minimal Arc
2	10	$1 + 4 \times 10^{-16}$	1.000	4.32×10^{-16}	4.06×10^{-16}	0.00
8	10	$1 + 4 \times 10^{-16}$	1.000	2.94×10^{-15}	2.54×10^{-15}	0.00
9	10	$1 + 2 \times 10^{-16}$	1.000	2.82×10^{-15}	2.23×10^{-15}	8.88×10^{-17}
10	10	$1 + 4 \times 10^{-16}$	1.000	3.59×10^{-15}	3.10×10^{-15}	8.88×10^{-17}
20	10	$1 + 2 \times 10^{-16}$	1.000	7.46×10^{-15}	6.22×10^{-15}	3.55×10^{-16}

The negligible numerical errors (within machine precision) and zero-phase spread conclusively demonstrate that every decomposed gate constructed from S is functionally equivalent to its source unitary. This establishes that the proposed universal set preserves complete unitary behavior across arbitrary Qudit dimensions, thus satisfying the definition of functional equivalence for universal gate decomposition.

3.5 Efficiency over Alternative Universal Gate Sets

To assess the efficiency of the proposed Reck-based decomposition against existing frameworks, we conducted a comparative gate-count analysis with the Li–Roberts–Yin

(LRY) decomposition method [16]. Both approaches decompose arbitrary unitary matrices into experimentally realizable sub-operations; however, the LRY method relies on fully controlled single-qubit operations, while Reck’s formulation uses embedded two-dimensional rotations and phase shifts, leading to a more compact representation for higher-dimensional Qudits.

For each randomly generated unitary matrix of order below 30, ten independent trials were performed. The average number of resulting elementary gates was recorded for both decomposition schemes. Table 3 summarizes the outcomes. The proposed method visibly exhibits a significantly lower gate count for higher folds, with the advantage growing rapidly as dimensionality increases. This confirms that Reck’s subspace-rotation approach scales more efficiently for large Qudit systems, offering a tangible reduction in synthesis complexity.

Table 3: Gate Count Comparison between Reck’s & Li–Roberts–Yin’s Decompositions

Qudit States	Trials Count	LRY Method (Avg. Gates)	Reck’s Method (Avg. Gates)
2	10	2.0	3.0
5	10	20.0	12.0
12	10	132.0	68.0
16	10	416.0	122.0
20	10	520.0	192.0
23	10	598.0	255.0
27	10	702.0	353.0

3.6 Validation over Quantum Cryptographic Models

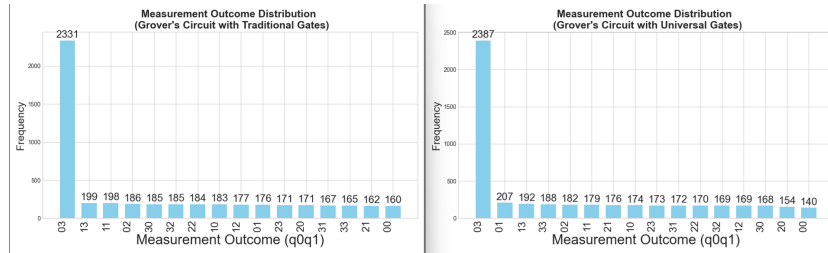
To establish the validity of our proposed minimal universal gate set in realistic scenarios, we performed detailed simulations of two cryptographically significant quantum models, Grover’s search algorithm and the QKD protocol. These models represent complementary aspects of quantum cryptology: the former serves as a benchmark for quantum search and cryptanalysis, while the latter underpins secure communication. In each case, we constructed two parallel circuits: one employing traditional Qudit gates, and another with universal gates synthesized through Reck’s decomposition.

The comparative study demonstrates that the proposed universal set faithfully reproduces the expected cryptographic behavior while providing a more generalizable and hardware-agnostic foundation. Importantly, the decomposed circuits illustrate how universal gate sets enable a consistent framework for both constructive protocols (QKD) and adversarial primitives (Grover’s search).

Grover’s Algorithm with Universal Decomposition: Grover’s search is a classic quantum algorithm, often studied for its cryptanalytic implications because of its known quadratic speed-up in unstructured search. In the multi-Qudit setting, it requires a generalization of the oracle and diffusion operators, both of which depend heavily on the availability of Hadamard-type gates for state initialization and inversion about the mean.

In the traditional construction, the circuit begins with two 4-dimensional Qudits, each initialized using explicit Qudit Hadamard operators. The oracle (U_f) flips the phase of the marked state, and the inversion operator (U_0) reflects amplitudes about the average. These transformations are expressed as dense, high-dimensional unitary matrices, directly invoked in the simulation. In the universal decomposition construction, we replaced the Hadamard gates with their Reck-decomposed equivalents. Each Hadamard operator was expressed as a product of two sets of matrices: a diagonal phase operator Φ , and a sequence of pairwise rotation matrices R_{ij} , embedded into the multi-Qudit Hilbert space. Thus, instead of a single dense Hadamard operation, the decomposed circuit applies multiple layers of R_{ij} -based rotations and Φ -phase shifts. This decomposition ensures that the gate belongs to the minimal universal set $\mathcal{S} = \text{PHASE}_1 \cup T_{\text{elements}}$, and thereby establishes universality.

Figure 2 demonstrates outcome of simulation of Grover’s Algorithms, where circuits have been created in two ways, using traditional Qudit gates and using proposed universal set of gates. While the simulation outcomes of both circuits were equivalent — amplifying the probability of the same marked state 03—the architectural distinction is fundamental. Traditional circuits rely on pre-defined, higher-level operators, whereas the decomposed version explicitly constructs the same effect using a basis of minimal gates. This reflects the central strength of the proposed framework: cryptographic models can be fully expressed using a universal and minimal gate set, eliminating the need for ad hoc gate assumptions tied to qubit-based universality. The figure illustrates that both constructions amplify the marked state with nearly identical distributions, affirming the correctness of the decomposition. Beyond correctness, this validates that cryptanalytic tasks can be generalized across dimensions while preserving efficiency. For cryptology, this means that Qudit-based Grover search can be universally synthesized and is not dependent on hardware-specific implementations of higher-level unitaries.



steps: Alice encodes random symbols in one of two bases, Bob independently measures in a chosen basis, and both parties compare basis choices to derive a shared secret key.

In traditional construction, Alice’s information encoding and Bob’s measurement were implemented directly with Qudit Hadamard and phase-type operators to generate rectilinear and diagonal bases. Bob’s measurement outcomes were obtained by applying the corresponding inverse basis transformations before projective measurement. In the universal decomposition construction, these basis-encoding and decoding operations were built entirely from Reck-decomposed gates. Each transformation, rather than being called as a predefined unitary, was explicitly realized as a composition of Φ -phase shifts and embedded R_i rotations. Alice’s encoded state and Bob’s chosen basis therefore originated from circuits constructed exclusively with universal gates, without reliance on traditional Hadamard-style operators. In our implementation, the two circuits were executed in parallel for 100 simulated rounds. In 47 cases, Alice and Bob selected the same basis, leading to the establishment of 47 secret key bits. In every such case, the key bits produced by the decomposed circuits matched those from the traditional implementation, confirming the semantic equivalence of the universal gates. The following two outcomes describe the operational differences between the QKD circuits with traditional and proposed universal gates.

```

--- QKD Round 29/100 ---
Alice: Preparing bit 2 in basis 'diagonal'.
Bob: Choosing basis 'diagonal'.
0(d=3): | Qutrit_Oswap2_Gate | Qu3H | Qu3H | M('diagonal_measurement') |
0(d=3): | Qu3M x3 | Qu3M x5 | Qu3M x5 | M('diagonal_measurement') |
Bob: Measured bit 2 in basis 'diagonal'.
Bob: Measured bit 2 in basis 'diagonal'.
Alice and Bob used the same basis.
Key established for this round: 2

```

In Round 29 of the QKD simulation, within the traditional setup, Alice randomly selected the trit “2” and the diagonal basis, while Bob independently chose the same diagonal basis. On Alice’s side, this resulted in a circuit beginning with the Oswap2 gate to encode the symbol, followed by a Hadamard gate to realize the diagonal basis. Bob, matching the basis choice, appended a Hadamard gate before measurement. In case of universal decomposed gates, all the gates are decomposed into unitary matrices of the proposed forms and then appended subsequently replacing the traditional gates. The Oswap2 gate has been replaced with one R_{ij} , one $\Phi_{balance}$ and one Φ matrices, while the Hadamard gate has been replaced with three R_{ij} , one $\Phi_{balance}$ and one Φ matrices. Both measurements yielded the value “2”, and since the bases aligned, a key bit was successfully established, exactly as predicted by the principles of QKD.

```

--- QKD Round 67/100 ---
Alice: Preparing bit 0 in basis 'rectilinear'.
Bob: Choosing basis 'diagonal'.
0(d=3): | Qu3H | M('rectilinear_measurement') |
0(d=3): | Qu3M | Qu3M | Qu3M | Qu3M | Qu3M | M('diagonal_measurement') |

```

```

Bob: Measured bit 1 in basis 'diagonal'.
Bob: Measured bit 2 in basis 'diagonal'.
Alice and Bob used different bases or measurement result invalid.
No key bit established for this round.

```

In Round 67 of the QKD simulation, within the traditional setup, Alice randomly selected the trit “0” and the rectilinear basis, while Bob independently chose the diagonal basis. Because of Alice’s choices, this resulted in a circuit that keeps the state unaltered. Bob, using his chosen basis, appended a Hadamard gate before measurement. In case of universal decomposed gates, the only Hadamard gate in the circuit has been replaced with three R_{ij} , one $\Phi_{balance}$ and one Φ matrices. We see both measurements yielded different values of measured bit, although the choice of bases being different, this round will anyway not yield any shared key bit. The simulation logs confirm consistent agreement between the universal and traditional circuits. This shows that decomposed universal gates not only preserve functional correctness, but also maintain the cryptographic guarantees of QKD protocols.

4 Conclusion

This work establishes the universality and minimality of the proposed gate set $S = PHASE_1 \cup T_{elements}$ for higher-dimensional Qudit systems. Minimality was rigorously proven using Lie algebraic arguments, confirming the indispensability of both phase and embedded $SU(2)$ transformations. A systematic decomposition algorithm was developed for arbitrary unitary transformations, analyzed for time and space complexity, and validated through functional and cryptographic experiments. Comprehensive evaluations confirm that the proposed decomposition preserves perfect functional equivalence across dimensions and achieves superior efficiency compared to existing universal gate constructions. Implementations on Grover’s search and QKD protocols further demonstrated correctness, resource transparency, and hardware independence, proving that high-level Qudit operations can be synthesized entirely from minimal universal primitives. The results reinforce three central implications for quantum cryptology: (i) universality in practice—validated through exact equivalence in decomposed circuits; (ii) hardware-agnostic portability—enabling scalable adoption across architectures; and (iii) efficiency and scalability—supporting faster QKD key generation and generalized Qudit-based search for cryptanalysis. Together, these findings present a unified, experimentally grounded framework where a single minimal universal gate set advances both defensive and adversarial dimensions of quantum cryptography. Future work will focus on physical realization and optimization of synthesis strategies to further minimize circuit depth and enhance scalability.

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